CSCI 5244: Homework Set 4

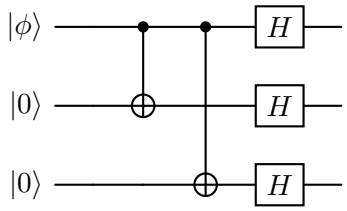
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1. **[Phase Flip Codes]** Recall the phase-flip error $\begin{cases} |0\rangle \to |0\rangle \\ |1\rangle \to -|1\rangle \end{cases}$, and the states $|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$ and $|1\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$. The following $[3, 1]$ codes can be used to detect and correct the phase-flip error:

$$\begin{cases} |0\rangle \to |\bar{0}\rangle = |+++\rangle \\ |1\rangle \to |\bar{1}\rangle = |---\rangle \end{cases}$$

   (a) Draw the quantum circuit for encoding the above phase-flip codes.

   Begin solution



   (b) Give the stabilizers that can be used to detect the phase-flip error.

   Begin solution

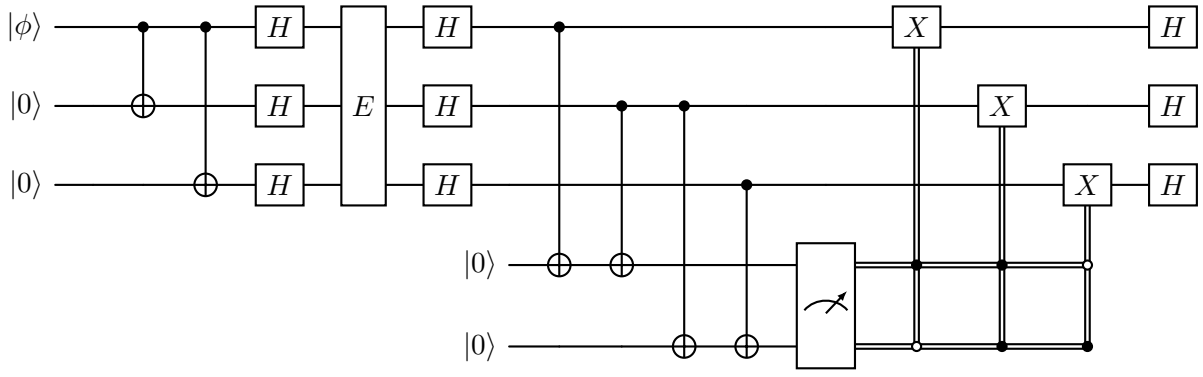
   Since this is a phase-flip error, we use the Pauli **X** matrices to flip $|+\rangle$ to $|-\rangle$.

   The stabilizers would be

$$M_1 = X_1 \otimes X_2 \otimes I$$
$$M_2 = I \otimes X_2 \otimes X_3$$

   (c) Continuing on the encoding circuit from (a), draw the circuit for error detection and correction.

   Begin solution



2. Recall Pauli X,Y,Z matrices: $\sigma_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \sigma_2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \sigma_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$

(a) Show the following anti-commutation relationships:

$$\{\sigma_i, \sigma_j\} = 2\delta_{ij} I$$

where δ_{ij} is the Kronecker delta.

Begin solution

An anti-commutator is a bilinear map defined as $\{X, Y\} = XY + YX$ and has the following properties

- $\{A, B\}^\dagger = \{B^\dagger, A^\dagger\}$
- $i\{A, B\}^\dagger = -iA, B$

(Taken from The Quantum Well)

The Kronecker delta is defined as: $\delta_{ij} = 1$ if $i = j$ else 0 if $i \neq j$.

To check these claims we can first check when $i = j = 1$ for $\{\sigma_1, \sigma_1\}$

$$\begin{aligned} \sigma_1\sigma_1 &= \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \\ \sigma_1\sigma_1 + \sigma_1\sigma_1 &= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} + \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \\ &= 2\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = 2I \end{aligned}$$

Next we can check for $\sigma_1\sigma_2$ which is

$$\begin{aligned} \sigma_1\sigma_2 &= \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix} = \begin{bmatrix} i & 0 \\ 0 & -i \end{bmatrix} \\ \sigma_2\sigma_1 &= \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} -i & 0 \\ 0 & i \end{bmatrix} \\ \sigma_1\sigma_2 + \sigma_1\sigma_1 &= \begin{bmatrix} i & 0 \\ 0 & -i \end{bmatrix} + \begin{bmatrix} -i & 0 \\ 0 & i \end{bmatrix} = 0 \end{aligned}$$

For σ_2 and σ_3

$$\begin{aligned} \sigma_2\sigma_3 &= \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} = \begin{bmatrix} 0 & i \\ i & 0 \end{bmatrix} \\ \sigma_3\sigma_2 &= \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix} = \begin{bmatrix} 0 & -i \\ -i & 0 \end{bmatrix} \\ \sigma_2\sigma_3 + \sigma_3\sigma_2 &= \begin{bmatrix} 0 & i \\ i & 0 \end{bmatrix} + \begin{bmatrix} 0 & -i \\ -i & 0 \end{bmatrix} = 0 \end{aligned}$$

Finally for σ_1 and σ_3

$$\sigma_1\sigma_3 = \begin{bmatrix}0 & 1\\ 1 & 0\end{bmatrix}\begin{bmatrix}1 & 0\\ 0 & -1\end{bmatrix} = \begin{bmatrix}0 & -1\\ 1 & 0\end{bmatrix}$$
$$\sigma_3\sigma_1 = \begin{bmatrix}1 & 0\\ 0 & -1\end{bmatrix} = \begin{bmatrix}0 & 1\\ 1 & 0\end{bmatrix}\begin{bmatrix}0 & 1\\ -1 & 0\end{bmatrix}$$
$$\sigma_1\sigma_3 + \sigma_3\sigma_1 = \begin{bmatrix}0 & -1\\ 1 & 0\end{bmatrix} + \begin{bmatrix}0 & 1\\ -1 & 0\end{bmatrix} = 0$$

(b) Show the following commutation relationships:

$$[\sigma_j, \sigma_k] = 2i\epsilon_{jkl}\sigma_l$$

where ϵ_{jkl} is the Levi-Civita symbol.

Begin solution

First, we can exapand $[\sigma_j, \sigma_k] = \sigma_j\sigma_k - \sigma_k\sigma_j$.

To check these claims we can first check when $j = 1, k = 2$. This gives us

$$2i(\epsilon_{121}\sigma_1 + \epsilon_{122}\sigma_2 + \epsilon_{123}\sigma_3) = 2i\sigma_3$$

$$\sigma_1\sigma_2 = \begin{bmatrix}0 & 1\\ 1 & 0\end{bmatrix}\begin{bmatrix}0 & -i\\ i & 0\end{bmatrix} = \begin{bmatrix}i & 0\\ 0 & -i\end{bmatrix}$$
$$\sigma_2\sigma_1 = \begin{bmatrix}0 & -i\\ i & 0\end{bmatrix}\begin{bmatrix}0 & 1\\ 1 & 0\end{bmatrix} = \begin{bmatrix}-i & 0\\ 0 & i\end{bmatrix}$$
$$\sigma_1\sigma_2 - \sigma_2\sigma_1 = \begin{bmatrix}i & 0\\ 0 & -i\end{bmatrix} - \begin{bmatrix}-i & 0\\ 0 & i\end{bmatrix} = 2i\begin{bmatrix}1 & 0\\ 0 & -1\end{bmatrix} = 2i\sigma_3$$

Check $j = 2, k = 3$

$$2i(\epsilon_{231}\sigma_1 + \epsilon_{232}\sigma_2 + \epsilon_{233}\sigma_3) = 2i\sigma_1$$

$$\sigma_2\sigma_3 = \begin{bmatrix}0 & -i\\ i & 0\end{bmatrix}\begin{bmatrix}1 & 0\\ 0 & -1\end{bmatrix} = \begin{bmatrix}0 & i\\ i & 0\end{bmatrix}$$
$$\sigma_3\sigma_2 = \begin{bmatrix}1 & 0\\ 0 & -1\end{bmatrix}\begin{bmatrix}0 & 1\\ 1 & 0\end{bmatrix} = \begin{bmatrix}0 & -i\\ -i & 0\end{bmatrix}$$
$$\sigma_2\sigma_3 - \sigma_3\sigma_2 = \begin{bmatrix}0 & i\\ i & 0\end{bmatrix} - \begin{bmatrix}0 & -i\\ -i & 0\end{bmatrix} = 2i\begin{bmatrix}0 & 1\\ 1 & 0\end{bmatrix} = 2i\sigma_1$$

Check $j = 3, k = 1$

$$2i(\epsilon_{311}\sigma_1 + \epsilon_{312}\sigma_2 + \epsilon_{313}\sigma_3) = 2i\sigma_2$$

$$\sigma_3\sigma_1 = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$$
$$\sigma_1\sigma_3 = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$$
$$\sigma_3\sigma_1 - \sigma_1\sigma_3 = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} - \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} = 2 \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} = 2i\sigma_2$$

For $j = k = 1$ we get

$$2i(\epsilon_{111}\sigma_1 + \epsilon_{112}\sigma_2 + \epsilon_{113}\sigma_3) = 0$$

$$\sigma_1\sigma_1 - \sigma_1\sigma_1 = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} - \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} = 0$$